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Early–Middle Devonian paleomagnetic results from the Zhongba microterrane, Tibetan Plateau: Evidence for its origin from the northern margin of Greater India

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ABSTRACT

The initial disintegration of Gondwana during the Paleozoic laid the foundation for the formation of the Tibetan Plateau in the Cenozoic. Determining the relative positions of the microterranes in Gondwana during the Paleozoic not only informs the subsequent drift and accretion processes of these microterranes but is also crucial to the paleogeographic reconstruction of Gondwana. However, the lack of paleomagnetic constraints on the Devonian paleogeography of the microterranes in the northern part of Gondwana makes this effort challenging. Here, we report paleomagnetic results for the first time from the Early–Middle Devonian sediments of the Zhongba microterrane. The site-mean direction is declination (*Ds*) = 310.7°, inclination (*Is*) = −67.2°, ks = 31.2, α_{95} = 8.3°, and *n* = 11, in stratigraphic coordinates. Positive fold and reversal tests, together with rock magnetism results and microscopic observations, strongly suggest that the remanence carriers are of depositional origin. The paleomagnetic results meet the paleomagnetic reliability criteria and therefore can be used for tectonic reconstructions. Our results constrain the paleolatitude of the Zhongba microterrane to be 50.0°S ± 11.7°S in the Early–Middle Devonian. Combined with published detrital zircon ages as well as paleomagnetic results and geological data, our data indicate that the Zhongba microterrane, which had no tectonic affinity with the Lhasa terrane, was part of the northern margin of Greater India during 408–388 Ma and coupled with the South Qiangtang terrane, Tethyan Himalayas, and other terranes to form the continuous northern continental margin of East Gondwana.

INTRODUCTION

As the highest plateau in the world, the Tibetan Plateau, known as the roof of the world, was formed by a series of amalgamations of microterranes originating from East Gondwana, involving a long northward drift and multistage convergence process (Fig. 1A; Metcalfe, 2013; Ding et al., 2014; Zhou et al., 2016; Cheng et al., 2023). These terranes of different natures and types, such as the North and South Qiangtang terranes, Lhasa terrane, Zhongba microterrane, etc., were important components of East Gondwana in the Paleozoic (Yin and Harrison, 2000), and they recorded the evolution of the Tethys Ocean and the breakup of East Gondwana. They also reshaped the topography of Asia during their subsequent aggregation and accretion, thus exerting far-reaching influences on the regional and global climate and environment changes (Wang et al., 2012; Metcalfe, 2013; Song et al., 2017; Ma et al., 2019; Guan et al., 2021; Cheng et al., 2022, 2023; Xiao et al., 2023; Yu et al., 2022; Wei et al., 2022, 2023). Therefore, reconstructing the Paleozoic paleogeography of these terranes and constraining their tectonic affinities with each other are crucial for understanding the paleogeographic pattern of East Gondwana and the evolution of the Tethys Ocean at that time.

As an important geological entity in the western part of the Yarlung Zangbo suture zone, the Zhongba microterrane is considered to be a narrow, small tectonostratigraphic terrane (Li et al., 2014a, 2014b, 2023; Zhang, 2015; He et al., 2016; Liu et al., 2019; Lin et al., 2020). Its continuous Paleozoic marine sedimentary sequence preserves important information on its tectonic affinity and provides an excellent geological record for studying the paleogeography of Gondwana (Li et al., 2014a, 2014b; Lin et al., 2020). However, the origin and attribution of the Zhongba microterrane in the Paleozoic is still a focus of debate. Whether the Zhongba microterrane was part of the southern margin of the Lhasa terrane (Xizang Autonomous Region Geology and Mineral Bureau, 1993; Zhang et al., 2007) or a component of the northern margin of India (Sun and Hu, 2012; Li et al., 2014a, 2014b, 2023; Lin et al., 2020) during the Paleozoic is still unclear, which seriously hinders our understanding and exploration of the tectonic evolution of Gondwana.

Paleomagnetism is one of the effective tools for quantitatively constraining the paleogeographic location of terranes in a certain period of geological history. The crustal terranes currently composing the Tibetan Plateau are geologically correlated to East Gondwana (Metcalfe, 2013), yet paleomagnetic data show a separated position for the North Qiangtang terrane in the Silurian–Permian (Song et al., 2017; Ma et al., 2019; Guan et al., 2021; Jiang et al., 2022; Cheng

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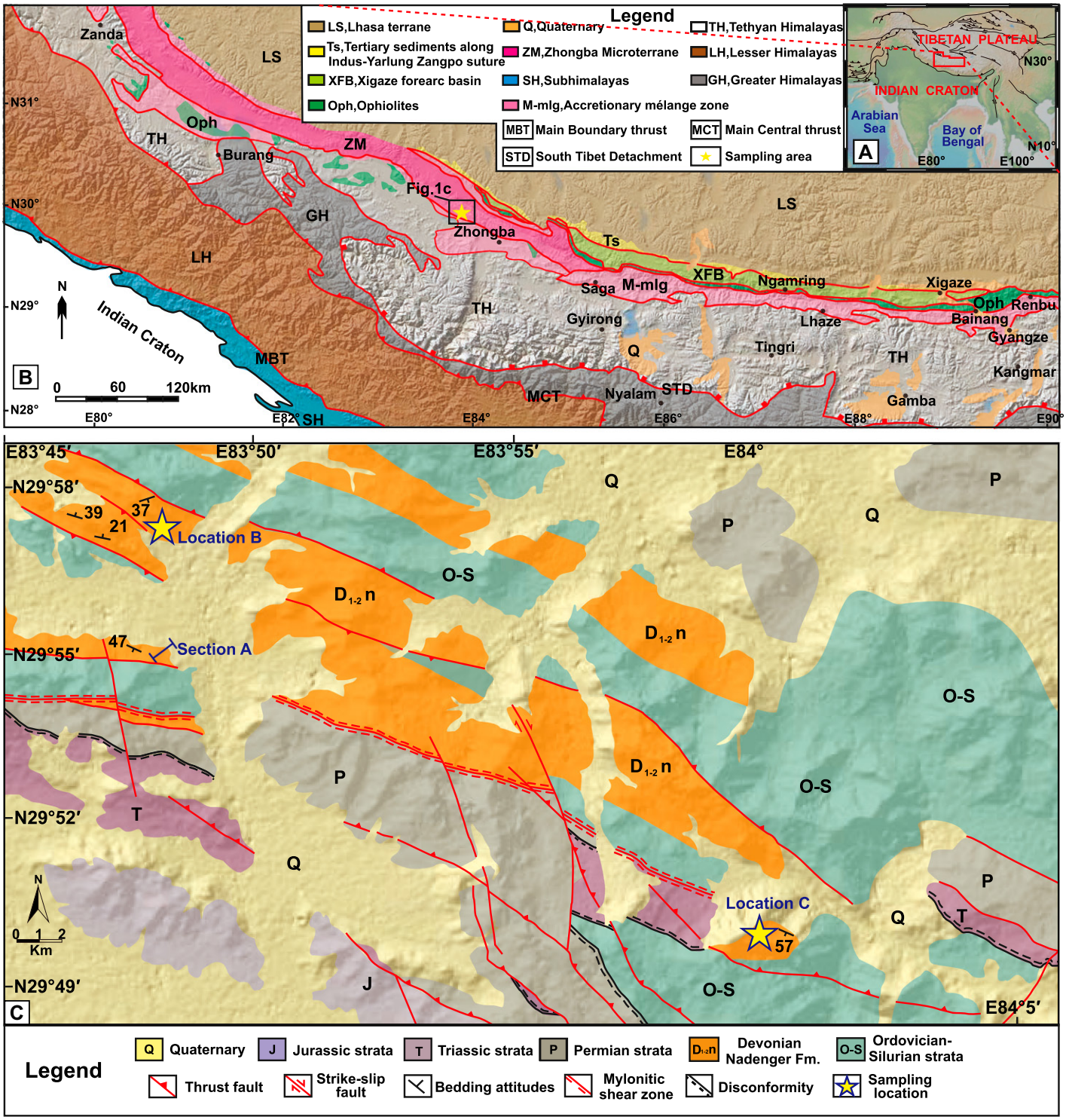


Figure 1. Regional distribution maps and paleomagnetic sampling locations in the Zhongba microterrane. (A) Topographic map of the Tibetan Plateau and the surrounding region. (B) Simplified geological map of southern Tibet and the Himalayan orogenic belt (modified from Wang et al., 2012; Hu et al., 2017) showing the main tectonic units and the distribution of the Zhongba microterrane. (C) Geological map of the northern part of Zhongba County showing the stratigraphic distribution and the paleomagnetic sampling locations (yellow star).

et al., 2023, and references therein). So far, there are few paleomagnetic data indicating the connection between the Tibetan terranes and Gondwana; particularly, paleomagnetic work in the Zhongba microterrane in the late Paleozoic is yet to be conducted.

In this paper, we report new paleomagnetic data from the Early–Middle Devonian Nadenger Formation of the Zhongba microterrane, aim-

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ing to precisely constrain the late Paleozoic paleogeography of the Zhongba microterrane. Combining published detrital zircon U-Pb age data, plate paleogeographic reconstruction, and paleontological records, we clarify the tectonic affinity of the Zhongba microterrane and revise the paleogeographic pattern of East Gondwana in the Devonian.

GEOLOGICAL SETTING, SAMPLING, AND EXPERIMENTAL METHODS

As the southernmost suture zone of the Tibetan Plateau, the Yarlung Zangbo suture zone is sandwiched between the Lhasa terrane to the north and the Tethyan Himalayas to the south (Fig. 1B), and it is a remnant of the Neo-Tethys Ocean that was preserved after undergoing tectonic deformation during the Cenozoic collision of the Indian continent with Eurasia. The Zhongba microterrane occupies a narrow area, ~1000 km long and 25 km wide, extending from Saga County in the southeast to the region near the China-India border (Fig. 1B). The Zhongba microterrane was previously regarded as part of the Tethyan Himalayas (Gansser, 1964) until Li et al. (2014a) defined it as an independent microterrane. It is composed of Sinian/Ordovician–Triassic marine strata of mainly low-grade metamorphosed crystalline limestone and sandstone (Li et al., 2014a, 2014b; He et al., 2016). It dismembers the ophiolite belt and accretionary wedge in the western section of the Yarlung Zangbo suture zone into two branches, the south and the north branches (Fig. 1B; Wang et al., 2012), and thus, the Zhongba microterrane was likely an independent crustal terrane before accretion to the Lhasa terrane or the Tethyan Himalayas. The two ophiolitic mélange zones are ~700 km long, with a maximum width of no more than 30 km, and they are composed of different tectonic blocks and matrixes. The southern mélange zone is relatively large in scale, while the northern mélange zone is sporadically distributed (Fig. 1B; Wang et al., 2012). Chronological studies indicate that the ages of the ophiolites within the two branches range from 130 Ma to 120 Ma (Dai et al., 2011).

The study area is located in the eastern part of the Zhongba microterrane, and the sampled strata are the Early–Middle Devonian Nadenger Formation. The Nadenger Formation strata are in fault contact with Permian strata, and its depositional environment was a neritic carbonate platform (Li et al., 2014a, 2014b). We collected a total of 160 core samples (11 paleomagnetic sampling sites) from the Nadenger Formation in the northern part of Zhongba County. Among them, 60, 58, and 42 samples were collected at location A (29°55′3.2″N, 83°48′48.4″E), location B (29°57′15.9″N, 83°47′52.3″E), and location C (29°50′16″N, 84°0′6.6″E), respectively (Fig. 1C). At locations A and C, the strike of the strata is in a SE-NW direction with a dip of 50°–60° (Fig. 2), while at location B, the strata dip toward the east with a dip range of 15°–40°. The main lithologies of the sampled strata are gray-yellow crystalline limestone and dark-gray medium-thick bioclastic limestone (Fig. 2). On one hand, algal and tentaculitoid clasts were found in the middle of the Nadenger Formation at locations A and B, respectively (Figs. 2A and 2B). Unfortunately, due to the poor preservation of the fossils, they cannot be used for identification. On the other hand, conodont fossils found in the upper part of the Nadenger Formation at location C were identified as *Icriodus* sp., *Ozarkodina* sp., *Polygnathus linguiformis*, *Polygnathus*, and *Pandorinellina* sp., indicating an age of Early–Middle Devonian (Li et al., 2014a, 2014b). Through structural analysis, microstructural observation, and Ar-Ar chronological research, Liu et al. (2019) argued that the deformation of the Zhongba microterrane under N-S compression was mainly divided into three stages: (1) the formation of large, broad, and gentle folds (Late Triassic to Early Cretaceous); (2) a stage of further deformation of the folds, during which there was a large amount of intralayer shear (Early Cretaceous–Late Cretaceous); and (3) the development of nearly E-W–trending, high-angle southward thrust faults (Late Cretaceous–Eocene). Therefore, we believe that the age of folding in the Nadenger Formation of the Zhongba microterrane was generally Cretaceous.

We used portable gasoline-powered drilling rigs to obtain structural paleomagnetic samples and then applied sun and magnetic compasses to orient the core samples. Subsequently, the samples were cut into standard cylindrical specimens for further experiments. All paleomagnetic demagnetization experiments were conducted at the South China Sea Institute of Oceanology, Chinese Academy of Sciences. The specimens were subjected to stepwise demagnetization with two Schonstedt TSD-1 type thermal demagnetizers and a 2G Model 600 alternating field (AF) demagnetizer, and their remanent magnetizations were measured with a 2G 755R superconducting rock magnetometer with an accuracy of $<5 \times 10^{-12}$ A·m^2 in a shielded room with a residual magnetic field intensity of less than 300 nT. The susceptibility versus temperature (*k-T*) experiment was carried out at the State Key Laboratory of Marine Geology, Tongji University, using an AGICO MFK1-FA. Isothermal remanent magnetization (IRM) experiments and microscopic observations of the thin sections were carried out at the China University of Geosciences (Beijing). We used an impulse magnetizer to gradually magnetize eight samples with an applied magnetic field ranging from 0.01 T to 2.5 T, and we used a dual-speed spinner magnetometer (JR-6A) to gradually measure the magnetization intensity of the samples. Then, we applied magnetic fields of 2.5 T, 0.4 T, and 0.12 T to the *x*, *y*, and *z* axes of the representative samples, respectively (Lowrie, 1990). Subsequently, a stepwise thermal demagnetization experiment was carried out. The maximum temperature of thermal demagnetization was 680 °C, measured with a superconducting magnetometer (2G 755R).

PaleoMac (Cogné, 2003) software and Paleomagnetism.org (Koymans et al., 2016) were applied to analyze the paleomagnetic data, including principal component analysis (Kirschvink, 1980), remagnetization great circles analysis (Halls, 1978; McFadden and McElhinny, 1988), Fisher mean calculation (Fisher, 1953), fold tests (e.g., McElhinny, 1964; McFadden, 1990), reversal tests (McFadden and McElhinny, 1990), and illustration of demagnetization data (Zijderveld, 1967). Cureval software (Hrouda, 1994) and Gplates (Williams et al., 2012) were used to present the *k-T* thermomagnetic runs (Petrovský and Kapička, 2006) and perform paleogeographic reconstructions, respectively.

RESULTS

Thermomagnetic Runs

The *k-T* curves of the Nadenger Formation samples showed significant changes during the heating and cooling processes in an argon atmosphere, especially in the temperature range of 400–580 °C (Figs. 3A–3D). The heating and cooling curves were nearly consistent within the interval of 450–580 °C, showing a significant decline to nearly zero. The pattern of *k-T* curves is typical for the Curie temperatures of magnetite (Dunlop and Ödemir, 1997). Moreover, the heating curves do not distinctly decrease in the temperature interval of 270–330 °C, indicating that the samples do not contain pyrrhotite (Dekkers, 1989). However, the cooling curves are much higher than the heating curves, which was likely caused by the oxidation of some iron-bearing minerals to form new magnetic minerals (Appel et al., 2012; Huang et al., 2017a, 2017b). To determine whether there is detrital magnetite in the Nadenger Formation samples, further microscopic observations are needed.

IRM Results

For the Nadenger Formation samples, two patterns of rising were observed in the IRM

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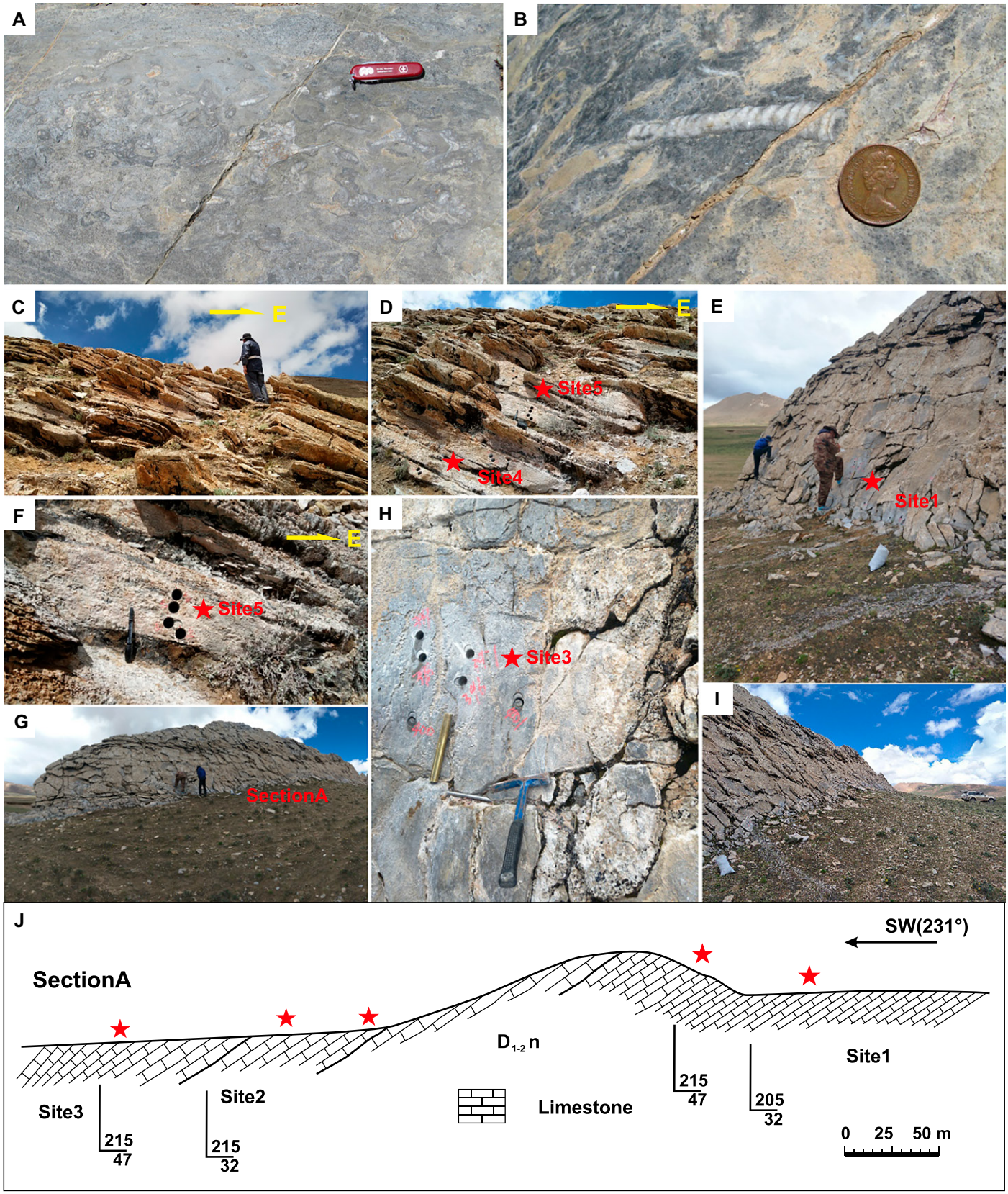


Figure 2. (A, B) Field photos of algal and tentaculitoid clasts in the middle of the Nadenger Formation (Fm). (C–I) Photos of field outcrops and paleomagnetic sampling sites (red star). (J) At location A, simplified geological cross section showing the approximate distribution of sampling sites.

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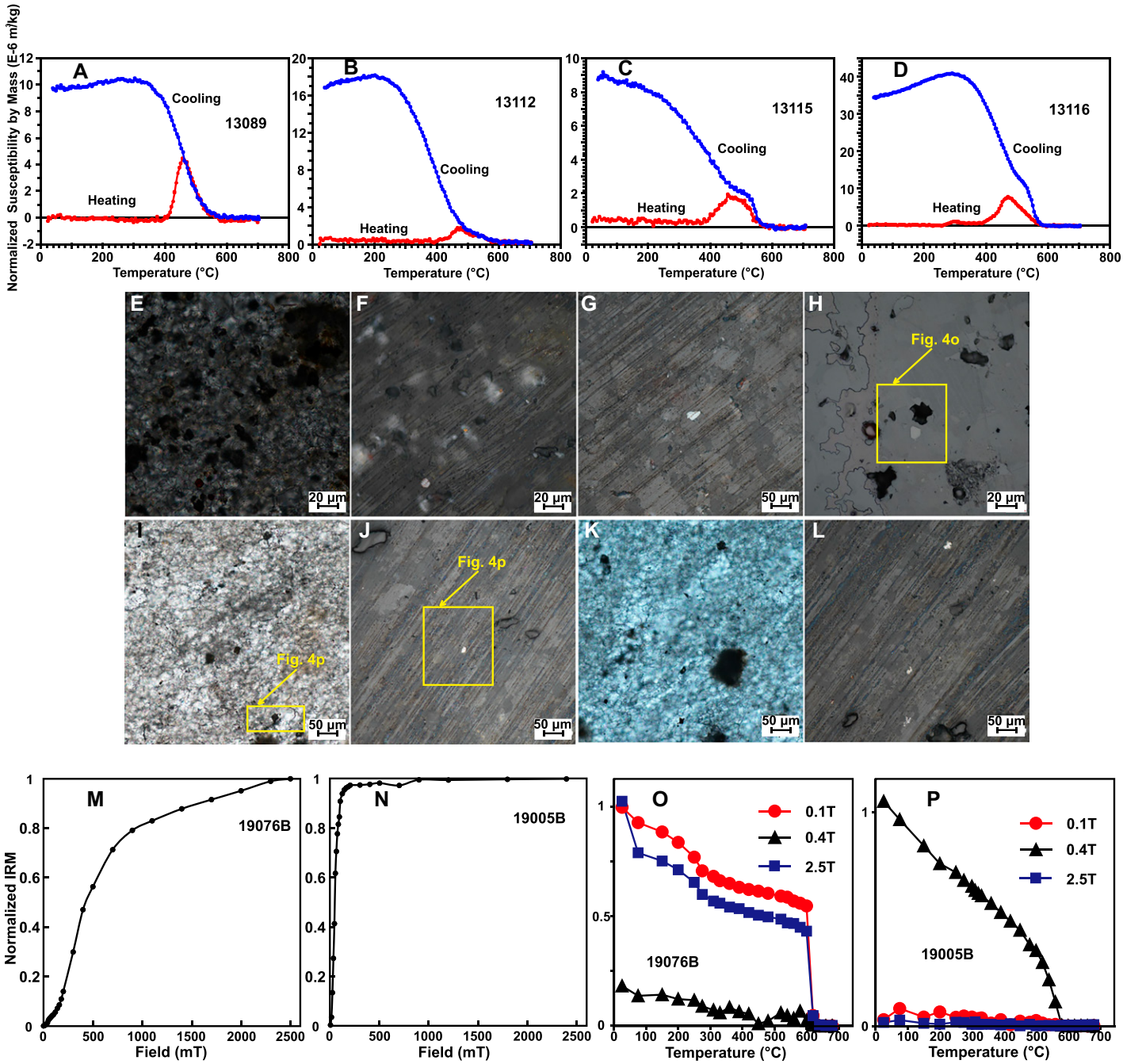


Figure 3. (A–D) Normalized magnetic susceptibility (*k*) versus temperature (*T*) curves. Blue and red curves indicate the cooling and heating processes, respectively. (E–L) Photomicrographs under an optical microscope. Photomicrographs E, I, and K were observed under single polarized light; the rest were taken under reflected light. Magnetite is steel gray under reflected light (F, G, H, J, L). The morphology of the metallic minerals in the yellow box can be seen more clearly in the scanning electron microscope images in Figure 4. (M–P) Normalized isothermal remanent magnetization (IRM) acquisition curves (M, N) and thermal demagnetization results of the orthogonal IRM components (O, P) for two representative samples. Red, black, and blue curves in the composite IRM thermal demagnetization experiments represent the soft, medium, and hard components, respectively (Lowrie, 1990).

acquisition curves. One is that when the intensity is less than 400 mT, the IRM acquisition curve rises rapidly as the magnetic field intensity increases. Subsequently, it keeps rising but does not reach saturation at 2500 mT, suggesting that the magnetic carriers are composed of both low- and high-coercivity minerals. Thermal demagnetization results of composite IRM data show that both the soft and hard components decrease sharply around 580–600 °C. The intensity of the medium component drops almost to zero around 470 °C. The hard component begins to decrease at 20–200 °C, indicating the presence of goethite with high coercivity (Lowrie, 1990). The other IRM acquisition curve pattern shows that when

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the magnetic field intensity is between 0 mT and 200 mT, the curve rises rapidly and reaches saturation at 200 mT, indicating that the magnetic carriers are low-coercivity minerals. Thermal demagnetization results of composite IRM data show that the medium component decreases sharply in the temperature range of 500–580 °C, which is consistent with the Curie temperature of magnetite (Dunlop and Ödemir, 1997). To sum up, the IRM experimental results show that the magnetic carrier in the Nadenger Formation samples is mainly magnetite.

Microscopic Observations

Thin sections were prepared for representative samples to perform a microscopic investigation. First, the metallic minerals in the thin sections were black and opaque under single polarized light (Fig. 3). Second, under reflected light, the reflection colors of the metallic minerals were steel gray (Fig. 3), which is consistent with the reflection color of magnetite. Third, to determine the element content and microscopic morphology of the minerals, we also carried out scanning electron microscope (SEM) observations and energy-dispersive spectrometry (EDS) analysis. The parameters were set as follows: The extra high tension (EHT) was 20 kV, and the working distance (WD) was 15 mm. A secondary electron detector (SE2) and angle selective backscatter detector (ASB) were used to observe the morphology and structure of the metallic minerals (Fig. 4). SEM observations and EDS analysis indicated that the mineral elements were mainly oxygen and iron, and no sulfur element was detected (Fig. 5). Moreover, observed iron oxides appeared irregular in shape without oxidation rims and did not grow in the boundaries and crystal cracks of other minerals (Fig. 4), as is typical of detrital magnetic minerals that are more likely to preserve primary remanence. It has been established that pyrrhotite and pyrite are very common remanence carriers in remagnetized limestone (Appel et al., 2012). However, the lack of sulfur, irregular shapes of the metallic minerals, the steel-gray reflection colors, and the mineral element compositions strongly suggest that the magnetic mineral is mainly detrital magnetite, which was not affected by secondary chemical alteration (Xu et al., 1998; Roberts et al., 2013). This is also supported by the *k-T* curves, magnetic coercivity, and unblocking temperature spectra revealed by progressive demagnetizations. However, the effects of remagnetization cannot be completely eliminated, based only on rock magnetism and microscopic observations. In the subsequent section, we will further discuss the reliability of paleomagnetic data.

Paleomagnetic Results

We carried out progressive thermal (160 samples) and AF demagnetization (13 samples) experiments on the Nadenger Formation samples to test the remanent component composition. Of these samples, 40 samples were discarded after removal of the viscous component at around 200 °C because the low intensity of magnetic remanence prevented subsequent analysis. In addition, except for 10 samples showing a single component, most of the samples exhibited two components (Fig. 6). On the one hand, the component of the demagnetization temperature between 20 °C and 150 °C was defined as the low-temperature component (LTC). On the other hand, we defined the component that decayed linearly toward the origin point in the temperature range between 150–330 °C and 450–600 °C as the high-temperature component (HTC) and then considered it to represent the characteristic remanent magnetization (ChRM) (Fig. 6). As for the result of AF demagnetization, 6 out of 13 samples yielded a well-defined ChRM, identifiable in at least four consecutive steps between 16 mT and 50 mT (Figs. 6E, 6G, and 6H). Moreover, the demagnetization curves of the six other samples did not pass through the origin but were approximately distributed in a great circle (Figs. 6I and 6J), so we used the remagnetization great circles for further analysis (Halls, 1978; McFadden and McElhinny, 1988).

The LTC directions were more clustered in geographic coordinates ($D_g = 2.6°$, $I_g = 47.7°$, $k_g = 13.6$, $\alpha_{95} = 3.6°$, $n = 124$) than in stratigraphic coordinates, where D indicates declination, I indicates inclination, k indicates precision parameter, and α_{95} indicates the radius of the 95% probability ellipse around the mean direction. (Figs. 7A and 7B). Moreover, the mean LTC direction was indistinguishable from the present geomagnetic field ($D = 0.8°$, $I = 47.1°$) of the study area and the recent geocentric axial dipole field ($D = 0°$, $I = 49.0°$), indicating that it is a viscous remanent magnetization. The mean in situ HTC direction was $D_g = 260.0°$, $I_g = -65.2°$, $k_g = 5.0$, $\alpha_{95} = 7.5°$, and $n = 87$, and the tilt-corrected direction was $D_s = 312.2°$, $I_s = -64.1°$, $k_s = 11.7$, $\alpha_{95} = 4.6°$, and $n = 87$ (Figs. 7C and 7D). In addition, the site-mean direction of the 11 sites for the Nadenger Formation samples was $D_g = 252.2°$, $I_g = -65.4°$, $k_g = 6.0$, and $\alpha_{95} = 20.3°$ in geographic coordinates (Fig. 7E), and $D_s = 312.6°$, $I_s = -67.2°$, $k_s = 22.3$, and $\alpha_{95} = 9.9°$ in stratigraphic coordinates.

It is worth noting, however, that the Nadenger Formation bedding attitudes indicate the presence of an inclined fold (bearing = 141.8°; dip = 19.2°). Consequently, we performed tilt correction using an inclined fold axis on the Nadenger Formation samples, and the results showed that the corrected direction is $D_s = 310.7°$, $I_s = -67.2°$, $k_s = 31.2$, $\alpha_{95} = 8.3°$, and $n = 11$ in stratigraphic coordinates (Fig. 7F), corresponding to a paleolatitude of 50.0°S ± 11.7°S at the reference point of 29.9°N, 83.9°E. The precision parameter k was significantly improved, so we used the corrected direction for the subsequent analysis. The results for the mean direction of the 11 sites are listed in Table 1.

To test the original nature of remanence, we conducted fold and reversal tests on the ChRM directions. First, the McElhinny (1964) fold test was positive. The critical values (2.12 and 2.94) were smaller than the ratio of the precision parameter ($k_s/k_g = 5.18$), indicating that the ChRM directions were acquired before folding. Second, the correlation test was also positive at 95% and 99% confidence intervals (McFadden, 1990). The test value in geographic coordinates ($\xi_{1,g} = 8.448$) was larger than the critical values (3.865 and 5.378), whereas the tilt-corrected test value ($\xi_{1,s}$) was equal to 1.492, suggesting that there is no correlation between bedding attitudes and the mean direction. Third, the incremental fold test showed that the optimum degree of untilting is 91.8% ± 8.3% (Fig. 8A; Watson and Enkin, 1993), which is consistent with the direction-correction (DC) fold test (Fig. 8B; Enkin, 2003), indicating that the remanence was acquired prior to folding. Moreover, the range of optimum percent unfolding was 72%–103% for the bootstrapping fold test (Fig. 8C; Tauxe and Watson, 1994; Tauxe et al., 2016). Finally, the angle ($\gamma = 12.2°$) between the normal direction ($D_{s,n} = 319.0°$, $I_{s,n} = -62.8°$, $k_{s,n} = 26.5$, $n_n = 6$) and the reversed direction ($D_{s,r} = 116.6°$, $I_{s,r} = 71.6°$, $k_{s,r} = 56.4$, $n_r = 5$) was smaller than the critical value (critical $\gamma = 15.9°$), revealing a C-level reversal classification (McFadden and McElhinny, 1990). Therefore, the positive fold and reversal tests indicate that the ChRM of the Nadenger Formation samples is primary.

DISCUSSION

Reliability of Paleomagnetic Data

In order to further assess the validity of our data, we evaluated our results according to the paleomagnetic quality criteria (Van der Voo, 1990; Deenen et al., 2011, 2014; Meert et al., 2020). First, based on the *N*-dependent A_{95} envelope provided by Deenen et al. (2011, 2014), we obtained a confidence interval for A_{95} ranging from 4.6° to 18.1°, while the value of A_{95} for this study was 11.7°, within this above interval, indicating that the paleosecular variation (PSV) of

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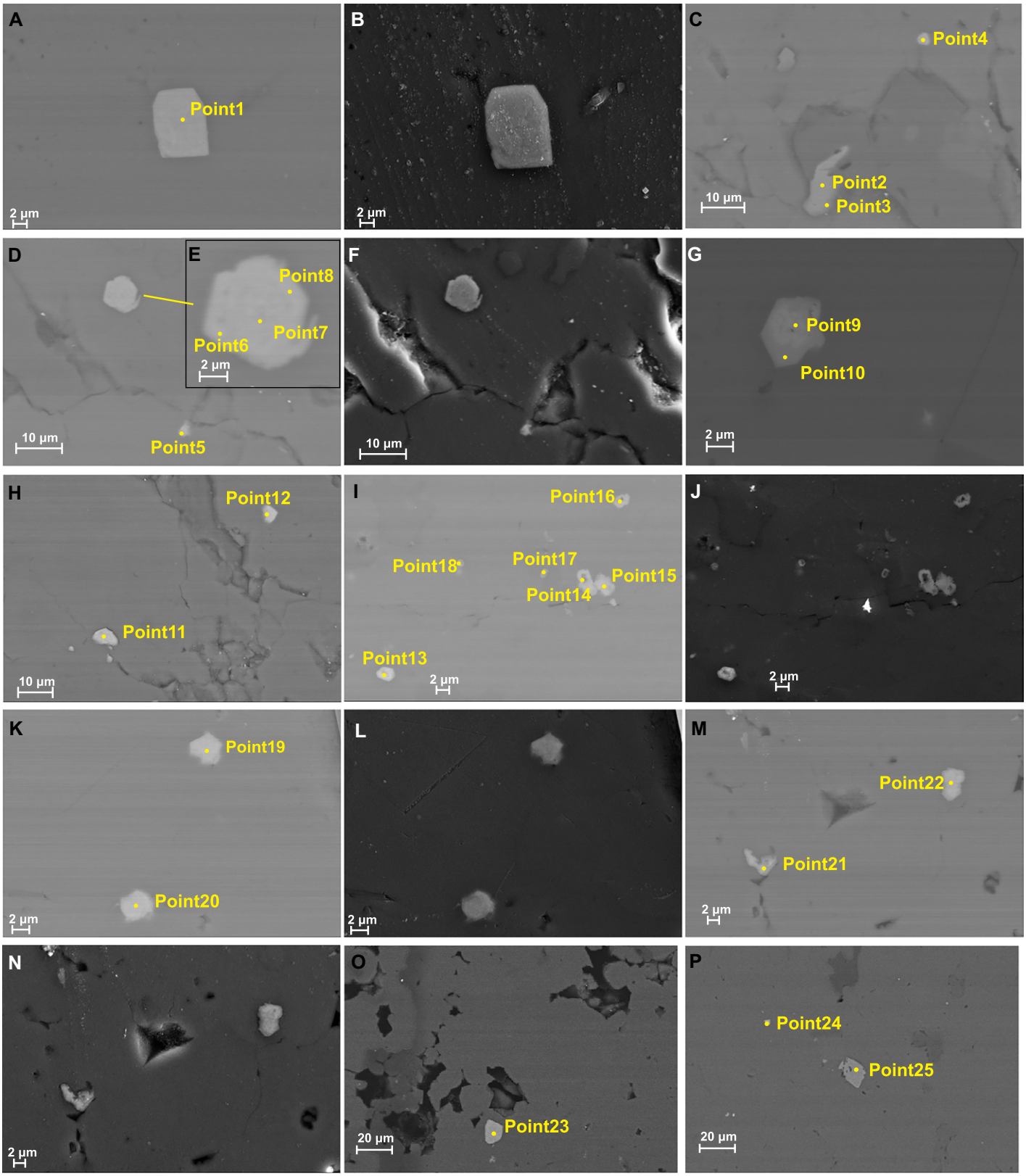


Figure 4. (A–P) Scanning electron microscope (SEM) observations. The morphology and structure of the metallic minerals in the Nadenger Formation samples were observed using the secondary electron detector (B, F, J, L, N) and the angle selective backscatter detector. The solid points (points 1–25) in the SEM images correspond one-to-one with the results of the energy-dispersive spectrometry (EDS) analysis.

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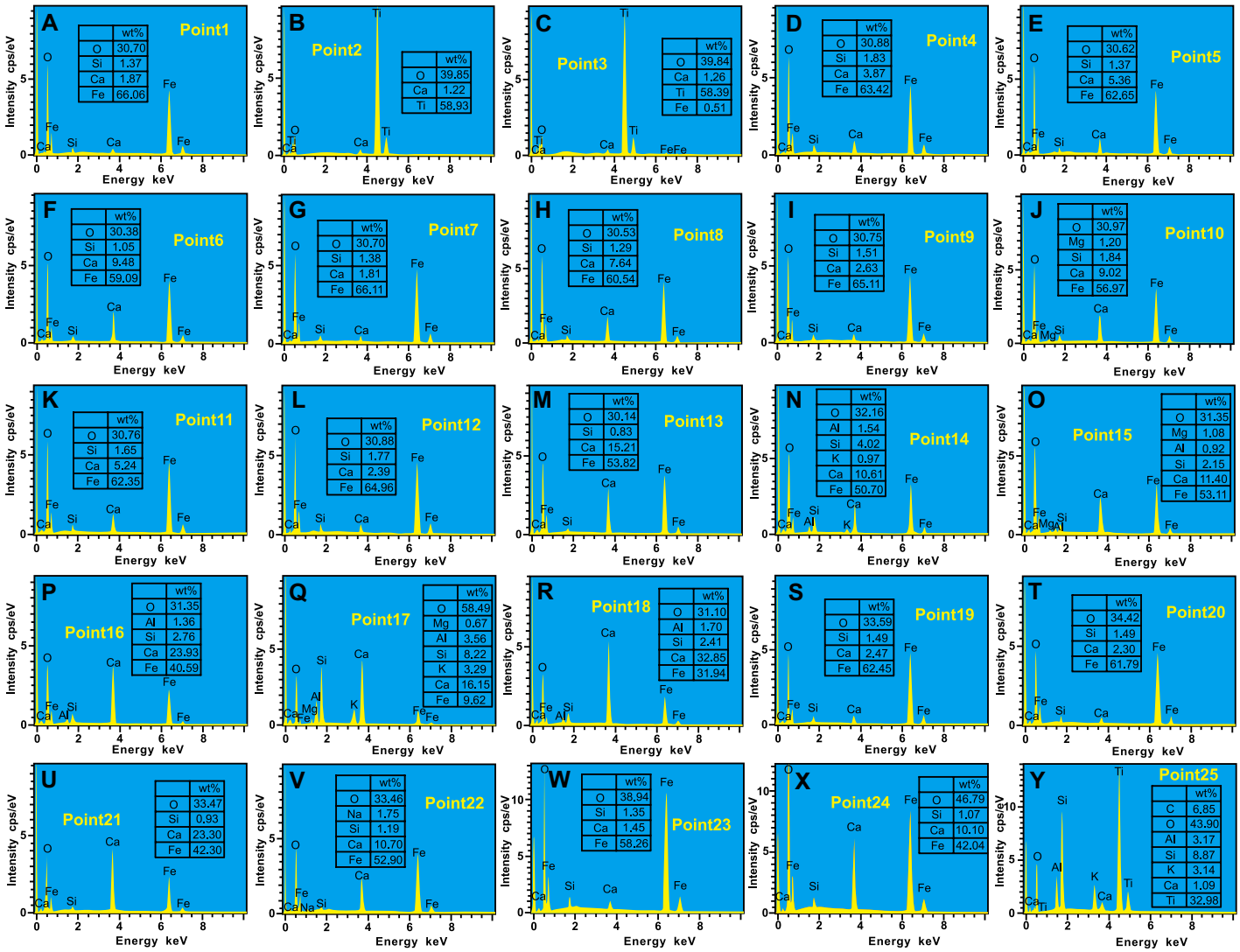


Figure 5. Energy-dispersive spectrometry (EDS) plots. The results show that the elemental composition of the magnetic carriers is mainly oxygen and iron. There are small amounts of titanium, magnesium, and aluminum (C, J, N–R, Y), probably representing isomorphic mixtures of iron oxides.

the geomagnetic field is averaged out. Additionally, the large variations of negative inclination (Fig. 7D) are likely the records of PSV in a normal polarity, which can be shown by the TK03 secular variation model (Tauxe and Kent, 2004).

Rock magnetism, petrographic observations, and EDS analysis are effective ways to eliminate remagnetization (Huang et al., 2017a, 2017b). In this study, positive fold and reversal tests, together with thermomagnetic runs, IRM results, and microscopic observations, strongly suggest that magnetite particles are of depositional origin. Our thermal demagnetization revealed an unblocking temperature of natural remanent magnetization (NRM) that is significantly lower than the Curie point (580 °C) of magnetite (Dunlop and Ödemir, 1997), but the unblocking temperature of IRM is consistent with the Curie point. Such observations can well be explained by the thermal viscous relaxation of NRM. In addition, due to the isomorphism of elements such as Mg, Ti, or Al in magnetite, the unblocking temperature of some Nadenger Formation samples may also be lower than the Curie temperature of magnetite (Fig. 5). A point worth noting is that our detailed microscopic observations showed that there are two particle sizes of magnetic carriers in the Nadenger Formation samples: One is 2–5 μm, and the other is 5–10 μm. The latter are larger particle remanence carriers and may also carry stable primary remanence (Zhao et al., 2021). Furthermore, remagnetization in the Tibetan Plateau, especially in the Tethyan Himalayas, is not widespread, and strata away from fluid migration may be immune to remagnetization (Xu et al., 2022), encouraging us to continue our attempts to obtain primary remanence, even in older strata.

The determined age of the sampled strata (Early–Middle Devonian) and application of multiple demagnetization techniques, coupled with positive fold and reversal tests, as well as detailed rock magnetism experiments and SEM observations, indicate that the paleomagnetic data reliability criteria (Q1–Q6) are met (Van der Voo, 1990; Meert et al., 2020). At present, no other Paleozoic paleomagnetic data from the Zhongba microterrane have been reported. However, geochemical data and paleontological evidence indicate that the Zhongba microterrane was located in a subtropical region (middle and low latitudes) of the Southern Hemisphere in the late Permian, separated from the Tethyan Himalayas, and

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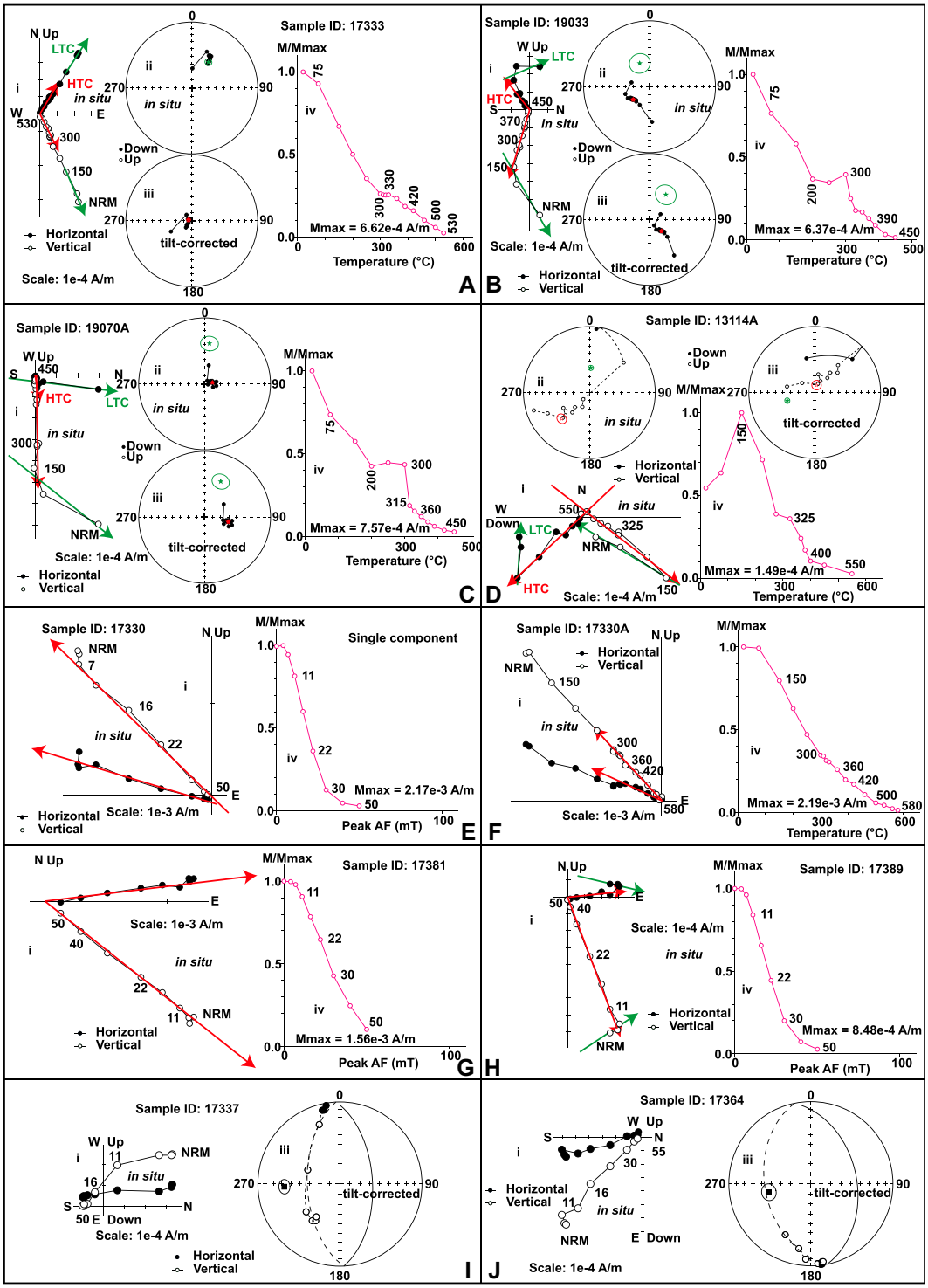


Figure 6. Diagrams showing the thermal (A–D, F) and alternating field (E, G–J) demagnetization results of typical samples from the Nadenger Formation. i—Zijderveld (1967) plots in the geographic coordinate system; ii and iii—equal-area projection plots in the geographic and stratigraphic coordinate systems, respectively; iv—normalized remanent intensity change graphs with heating temperature. Green and red arrows represent the low-temperature component (LTC) and high-temperature component (HTC) results, respectively, obtained by the principal component analysis method (Kirschvink, 1980). Parts I and J indicate remanence analyzed by the remagnetization great circles (Halls, 1978; McFadden and McElhinny, 1988). Green (red) stars represent the mean directions of the LTC (HTC). Circle associated with the star refers to its 95% confidence interval. Parts E and F are two specimens cut from the same core sample. Their thermal and alternating field (AF) demagnetization curves show consistent magnetic characteristics. NRM—natural remanent magnetization.

became an independent microterrane (Cheng et al., 2022; Xiao et al., 2023). Considering the current latitude, the Zhongba microterrane drifted from the Southern Hemisphere to the Northern Hemisphere as a whole. In this study, the ancient remanence indicates a high southern paleolatitude position for the Zhongba microterrane during the Devonian. If the magnetization was substantially delayed, it would show a lower Southern Hemisphere or even Northern Hemisphere latitude, and thus, our results also meet criterion Q7. Veikkolainen et al. (2014) considered that the reliability criterion Q7 should be discarded because it may wrongly define the primary remanence as being affected by remagnetization. This may be true when there was no significant plate-tectonic movement, but this is not applicable for the Zhongba microterrane.

In conclusion, the Early to Middle Devonian Nadenger Formation paleomagnetic results meet all seven paleomagnetic reliability criteria and can be used for tectonic paleogeographic reconstructions.

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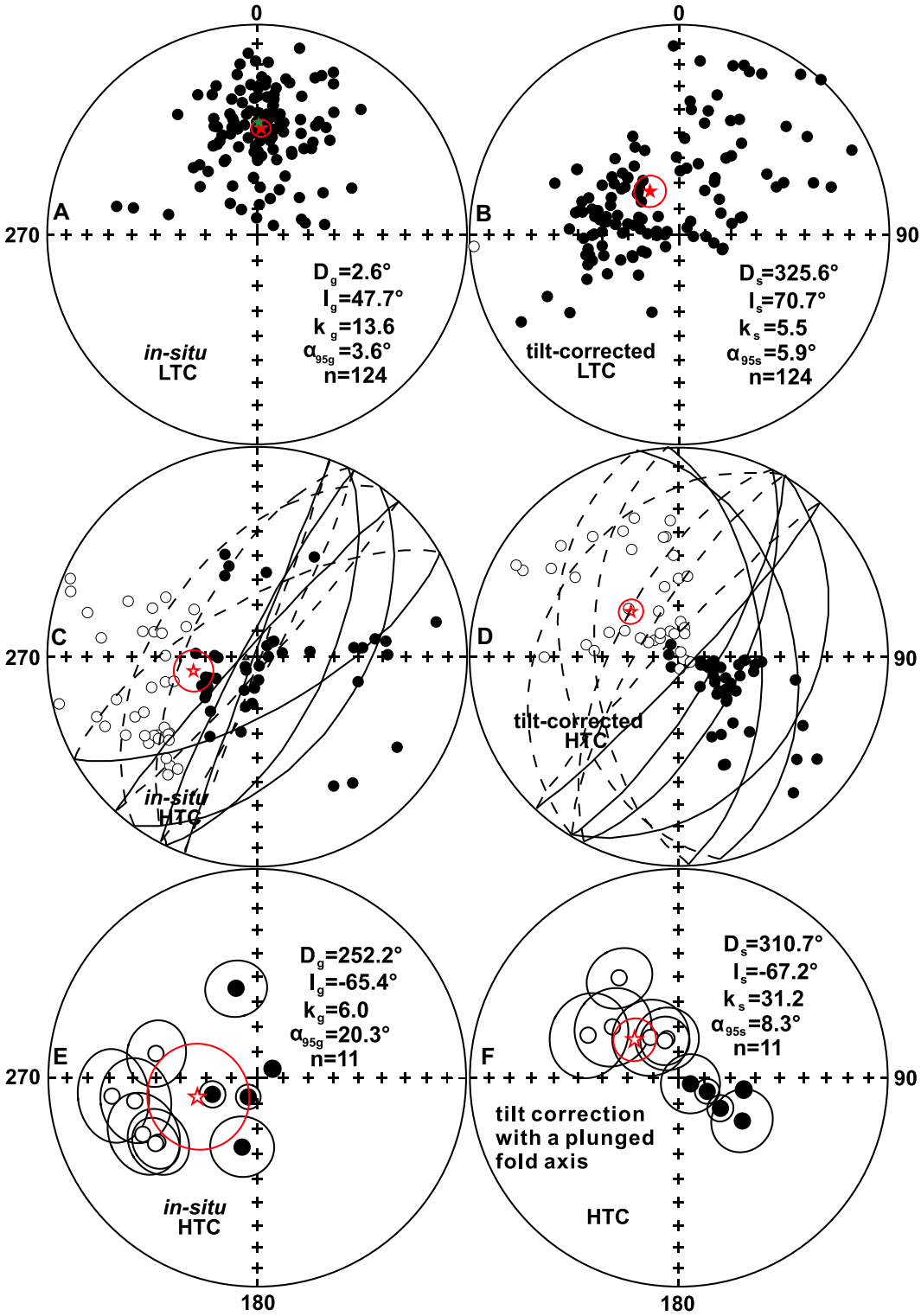


Figure 7. Equal-area projections of the low-temperature component (LTC; A, B), characteristic remanent magnetization (ChRM; C, D), and site-mean directions (E, F). Green and red stars represent the current geomagnetic field direction and the mean directions of the remanent components, respectively. Red circles indicate its 95% confidence interval. Solid and open dots refer to the projection of the lower and upper hemispheres, respectively.

Tectonic Affinity of the Zhongba Microterrane

The research on the tectonic affinity of the Zhongba microterrane is still controversial. One view holds that the Zhongba microterrane has tectonic affinity with the Lhasa terrane. The Zhongba microterrane might be a discrete terrane rifted from the southern margin of the Lhasa terrane in the early Hercynian–Indosinian period (Xizang Autonomous Region Geology and Mineral Bureau, 1993) or an accretionary wedge belonging to the Lhasa terrane (Zhang et al., 2007). Another view is that the Zhongba microterrane should be an independent terrane, which has different plate-tectonic affinity from the Lhasa terrane (Sun and Hu, 2012; Li et al., 2014a, 2014b, 2023; Lin et al., 2020).

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TABLE 1. SITE-MEAN CHARACTERISTIC REMANENT MAGNETIZATION (ChRM) DIRECTIONS OF THE EARLY–MIDDLE DEVONIAN SAMPLES FROM THE ZHONGBA MICROTERRANE IN SOUTHERN TIBET

Site	Strike/dip (°)	*n*/*N*	*Dg* (°)	*Ig* (°)	*Ds* (°)	*Is* (°)	*k*	α_{95} (°)
Site 1	125/47	11/16	259.3	−40.1	311.3	−53.5	10.1	15.1
Site 2	125/47	11/18	262.8	−30.6	300.3	−44.0	9.3	15.9
Site 3	125/47	9/15	283.5	−48.7	331.9	−39.9	18.7	12.2
Site 4	338/37	11/16	12.3	−62.2	120.2	60.6	15.3	12.1
Site 5	7/35	11/18	69.7	−71.4	121.9	72.7	81.0	5.1
Site 6	350/15	6/6	25.1	−81.8	112.1	77.5	192.0	4.8
Site 7	21/21	5/6	239.5	−83.1	98.6	64.1	562.1	3.2
Site 8	71/40	4/6	166.7	−53.7	106.4	84.9	65.0	11.5
Site 9	131/57	6/9	237	−42.7	352	−74.4	30.6	12.3
Site 10	131/57	8/14	243.7	−38.8	333.8	−70.9	14.8	14.9
Site 11	131/57	5/9	237.1	−41.6	358	−78.3	65.6	9.5
Mean corrected direction		87/133	252.2	−65.4			6.0	20.3
					312.6	−67.2	22.3	9.9
Tilt correction with a plunged fold axis					310.7	−67.2	31.2	8.3
						$A_{95} = 11.7$; $\lambda = 50.0°S$		

Note: n/N—number of samples used to calculate the mean directions and number of all samples analyzed; *Dg* (*Ds*)—declination in geographic (stratigraphic) coordinates; *Ig* (*Is*)—inclination in geographic (stratigraphic) coordinates; *k*—precision parameter; A_{95} and α_{95}—radius of 95% confidence circle about the pole and radius of the 95% probability ellipse around the mean direction; λ—paleolatitude.

To determine the tectonic affinity of the Zhongba microterrane in the Early–Middle Devonian, and to help quantify its relative position to the surrounding terranes, we compared the paleolatitude of the Zhongba microterrane during 408–388 Ma (50.0°S ± 11.7°S) with the expected paleolatitudes of India (Torsvik et al., 2012; van Hinsbergen et al., 2012) at 400–360 Ma for the reference point (29.9°N, 83.9°E; Fig. 9). We also collected paleomagnetic data from the Silurian to Carboniferous sections of the North Qiangtang terrane, Indochina terrane, and South China block. Our observed Early–Middle Devonian paleolatitude is consistent with the expected paleolatitudes from the Indian apparent polar wander path (APWP), and the paleomagnetic poles are also essentially close together (Fig. 9), indicating that the Zhongba microterrane was likely part of the northern margin of Greater India in the Early–Middle Devonian.

Our paleomagnetic inference for the Indian connection of the Zhongba microterrane is also supported by detrital zircon U-Pb age spectra of clastic rocks from Silurian–Carboniferous sedimentary strata (Lin et al., 2020) and the Devonian Nadenger Formation (Zhang, 2015) in the Zhongba microterrane. Lin et al. (2020) and Zhang (2015) obtained 168 and 449 valid data points of detrital zircon U-Pb ages, respectively. We compared the detrital zircon age data obtained from the Zhongba microterrane with the detrital zircon age information from the South Qiangtang terrane, Lhasa terrane, and Tethyan Himalayas (Fig. 10; Gao et al., 2022, and references therein). Moreover, based on the study of detrital zircon U-Pb age spectra and Hf isotopic composition from clastic rocks of the Permian Gangzhutang Formation in the Zhongba area, the Zhongba microterrane, South Qiangtang terrane, and Tethyan Himalayas contain typical age peaks of 530 Ma, 950 Ma, and 2500 Ma, similar $\varepsilon_{Hf}(t)$ values, and similar crustal model ages (950 Ma; Sun and Hu, 2012). In general, the distribution characteristics of the detrital zircon ages of the Zhongba microterrane are similar to those of the South Qiangtang terrane and Tethyan Himalayas, with a characteristic peak at ca. 950 Ma, which is different from the Lhasa terrane and Australia, which have a typical detrital zircon age of ca. 1170 Ma (Fig. 10; Zhu et al., 2011, 2013). As for the Lhasa terrane, Zhu et al. (2011) believed that its provenance was different from the Tethyan Himalayas, and it was likely connected with southwestern Australia.

Torsvik et al. (2009) conducted a paleomagnetic study on the Early Ordovician red beds (Shian Formation) of the Tethyan Himalayas, and the results support the idea that the Tethyan Himalayas were located in proximity to the Indian craton in the Paleozoic (within the error range), which is consistent with faunal distributions, stratigraphic correlations, and provenance age estimates (Myrow et al., 2009, 2010). Therefore, based on the comparison of the detrital zircon age peaks and paleomagnetic data, the Zhongba microterrane is interpreted to have belonged to the Tethyan Himalayas–Greater India–South Qiangtang tectonic system in the late Paleozoic, which together constituted the continuous northern margin of the Indian continent, but it had no plate-tectonic affinity with the Lhasa terrane (Sun and Hu, 2012; Li et al., 2014a, 2014b, 2023; Lin et al., 2020).

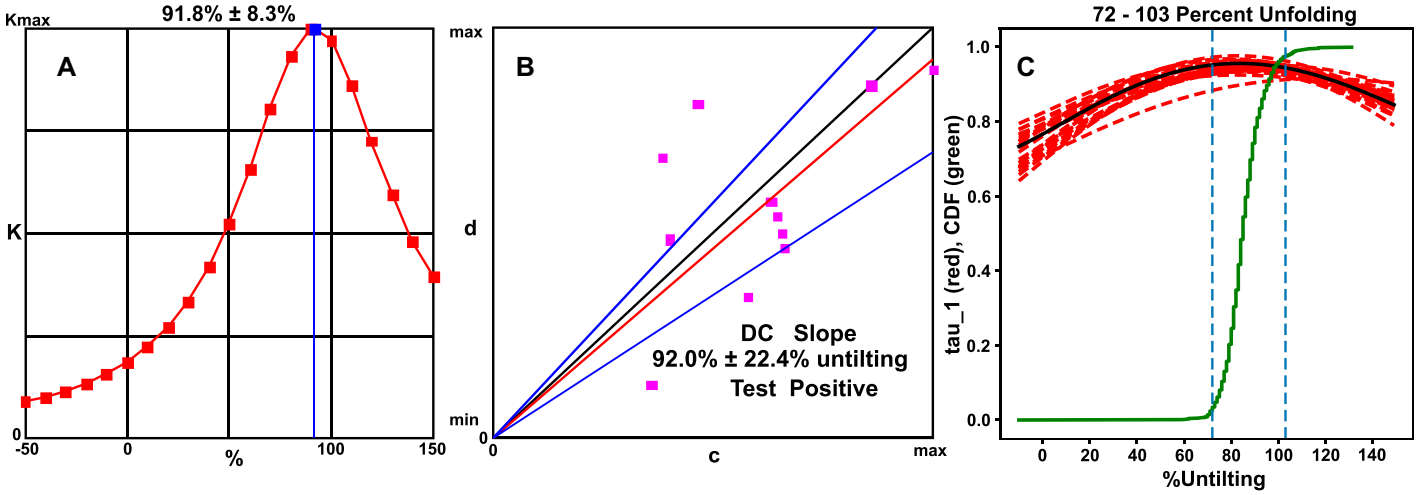


Figure 8. (A) Results of partial untilting test, (B) results of direction-correction (DC) fold test, and (C) results of bootstrapping fold test. K—precision parameter; CDF—cumulative density function.

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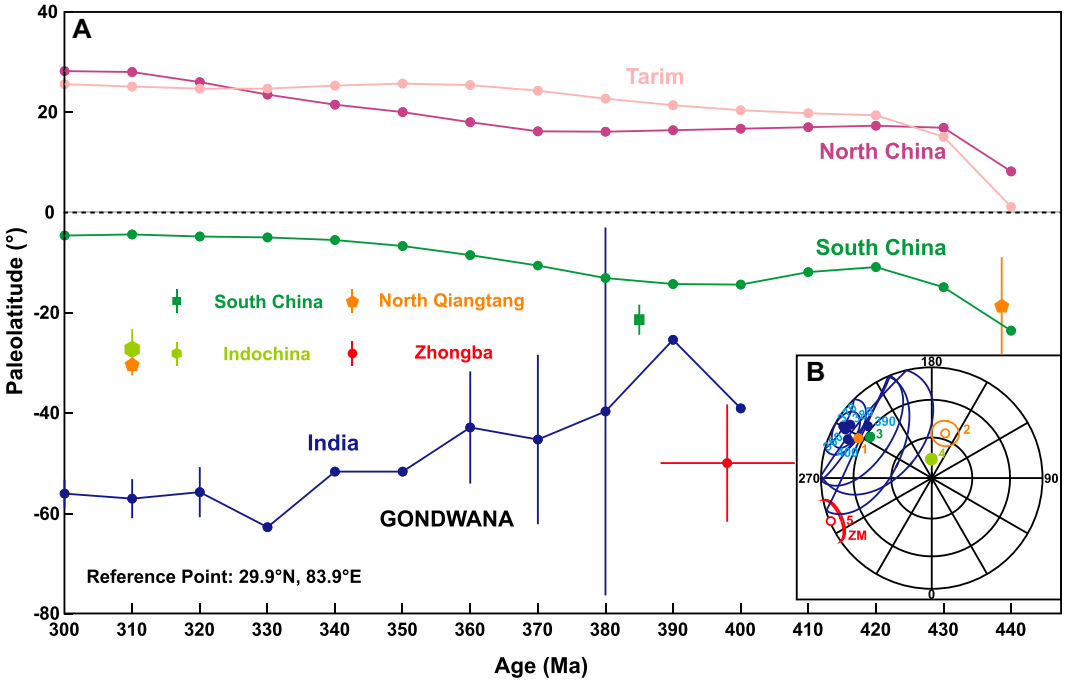


Figure 9. (A) Plots of paleolatitudes for the Zhongba microterrane and its surrounding terranes (South and North China blocks, North Qiangtang terrane, and Indochina terrane) and India during the Silurian to Carboniferous. Red dot indicates the result of this study. The horizontal and vertical lines represent the stratigraphic age and the error of paleolatitude, respectively. The reference point is 29.9°N, 83.9°E. The data are stored in the Supplemental Material (see text footnote 1). (B) Equal-area projection of the late Paleozoic paleomagnetic poles of the Zhongba microterrane and surrounding terranes and the paleomagnetic pole of India during 400–360 Ma. Red dot and associated ellipse represent the paleomagnetic pole and confidence interval of the Zhongba microterrane (ZM). The blue dots represent the paleomagnetic poles of India at different times. The other colored dots represent the paleomagnetic poles of the terranes surrounding the Zhongba microterrane. The paleomagnetic poles represented by the numbers can be seen in the Supplemental Material (Table S4 [see text footnote 1]).

Early–Middle Devonian Gondwana Paleogeographic Pattern

Recently, Wang et al. (2021) redefined Gondwana as a supercontinent—a continuous and long-existing continent formed by the collection of multiple terranes, usually divided into West Gondwana and East Gondwana. West Gondwana included the South America and Africa blocks, and East Gondwana included Sri Lanka, East Antarctica, Madagascar, India, Australia, and the Lhasa, Qiangtang, Indochina, and Sibumasu terranes (Zhao et al., 2018; Wang et al., 2021). The blocks or terranes with tectonic affinity to East Gondwana gradually fragmented in the Southern Hemisphere during the Paleozoic with the opening of the Paleo-Tethys and Neo-Tethys Oceans, and then they rejoined into the Eurasian continent in the Northern Hemisphere accompanied by the closure of the Tethys Oceans in the Mesozoic and Cenozoic, which provided the necessary conditions for the formation of the Tibetan Plateau during the Cenozoic (Metcalfe, 2013; Ding et al., 2014; Zhao et al., 2018; Wang et al., 2021). As the predecessor of the Neo-Tethys Ocean, the Paleo-Tethys Ocean is divided into the East Paleo-Tethys Ocean and the West Paleo-Tethys Ocean (Zhao et al., 2018). It is generally believed that it began in the early Paleozoic and closed in the Triassic, but the evolution of the East and West Paleo-Tethys Oceans and their branches may have been diachronous (Metcalfe, 2013; Stampfli et al., 2013; Zhao et al., 2018). This is also supported by new early–middle Silurian paleomagnetic data and geological evidence (such as the early Paleozoic ophiolite in the Longmuco-Shuanghu suture zone) from the North Qiangtang terrane (Zhai et al., 2016; Jiang et al., 2022). Jiang et al. (2022) reported primary paleomagnetic results from the early–middle Silurian Longmuco Formation and proposed that the North Qiangtang terrane was located in the low latitudes of the Southern Hemisphere (~20°), which may have been close to the South China block (23.6°S) at this time (Fig. 9; Huang et al., 2018; Zhao et al., 2018). Furthermore, Yang et al. (2017) constrained the North Qiangtang terrane to be located at 30.3°S in the late Carboniferous, while the middle Permian paleolatitude of the North Qiangtang terrane was 24°S (Cheng et al., 2023), so its stable paleolatitudinal position may have lasted from the Silurian to the middle Permian. These paleomagnetic data show that the North Qiangtang terrane was separated from Gondwana in the early Paleozoic. In other words, there was an ocean between them during the early Paleozoic. Moreover, Zhai et al. (2016) discovered the oldest ophiolite in the Gangma Co region of Tibet and proposed that the Paleo-Tethys Ocean opened in the early Paleozoic and closed in the Triassic. Recently, Xian et al. (2019) carried out detailed paleomagnetic work on the Devonian red beds in the South China block and combined their results with published paleomagnetic data to establish the APWP of the South China block. Then, they compared it with the APWP of Gondwana at the same time. They found that the East Paleo-Tethys Ocean between Gondwana and the South China block reached a width of 1600 km in the late Paleozoic, which is also supported by radiolarian fossil assemblages and geochemical evidence from Devonian siliceous rocks in northern Thailand (Hara et al., 2010). In summary, these pieces of evidence strongly support the interpretation that the East Paleo-Tethys Ocean likely opened in the late Paleozoic or before.

The Indochina terrane was located at 27.3°S ± 4.0°S in the late Carboniferous (Yan et al., 2020) and at ~30°N in the Late Triassic (Yan et al., 2017), showing a period of rapid northward drift. In the late Paleozoic, the Indochina terrane and the North Qiangtang terrane were consistent in paleolatitude (Fig. 9; Yang et al., 2017; Yan et al., 2020), and the two may have formed a ribbon-like continental archipelago (Cheng et al., 2023) and drifted northward

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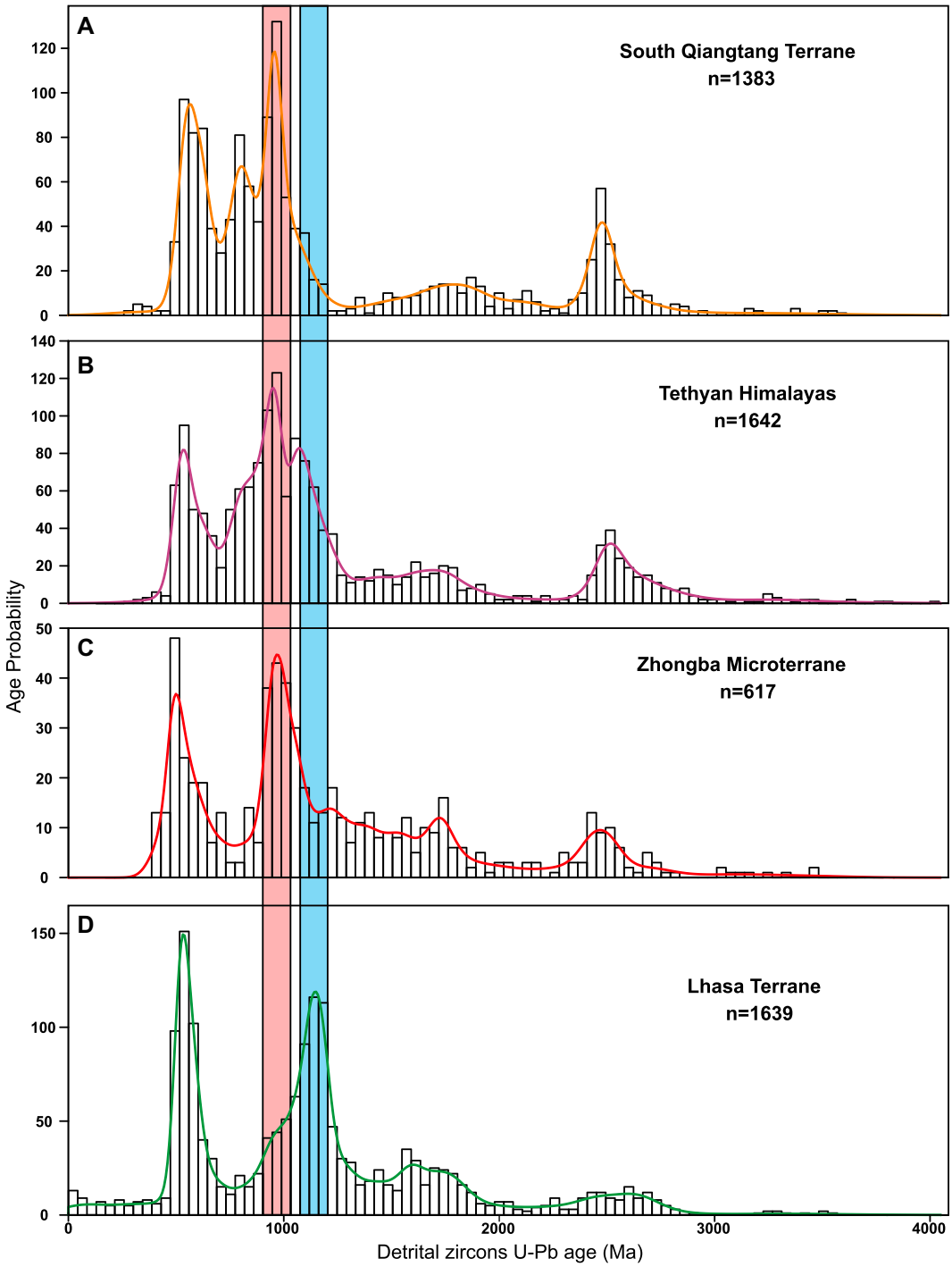


Figure 10. U-Pb age spectra of detrital zircons from the Zhongba microterrane (Zhang, 2015; Lin et al., 2020), South Qiangtang terrane, Tethyan Himalayas, and Lhasa terrane (Gao et al., 2022, and references therein).

together in subsequent movements. In addition, during the Silurian to Devonian periods, the Indochina terrane and the South China block have similar fossil vertebrate records (Wang et al., 2010). Therefore, we place the Indochina terrane, North Qiangtang terrane, and South China block together as a whole in the paleogeographic reconstruction of East Gondwana (Fig. 11). Moreover, during the late Paleozoic, the Tarim craton and North China blocks were located in the midlatitudes of the Northern Hemisphere (Figs. 9 and 11). There are no Devonian data from the Lhasa terrane, and in fact, we report the first Devonian dataset from Tibet, which shows a consistent paleolatitude with northern East Gondwana. In contrast, detrital zircon ages have confirmed that the Lhasa and Sibumasu terranes had tectonic affinities with Australia during the Paleozoic (Zhu et al., 2011, 2013; Dew et al., 2021; Gao et al., 2022), and therefore, we place them in northwestern Australia (Fig. 11). Likewise, based on detrital zircon data, we believe that the Zhongba microterrane, the Qiangtang terrane, and the Tethyan Himalayas were integrated with and located on the northern margin of East Gondwana, at least in the Early–Middle Devonian. This is also supported by paleobiogeographic divisions. Zhang et al. (2013) integrated Permian biostratigraphy and faunal compositions of the main

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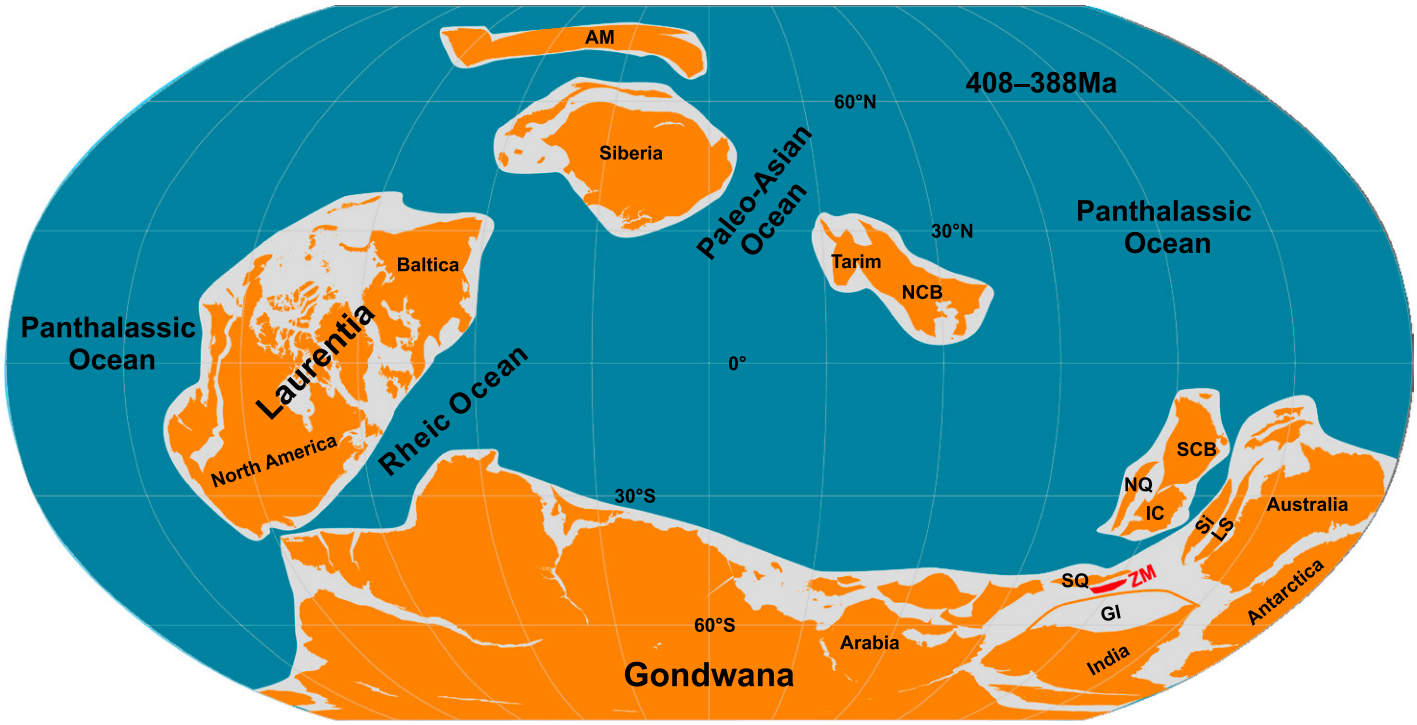


Figure 11. Global paleogeographic reconstruction during 408–388 Ma interval in Robinson projection generated by using the GPlates software (Williams et al., 2012). Red area indicates the Zhongba microterrane. Reconstruction map is modified from Torsvik et al. (2014) and Matthews et al. (2016), where the Lhasa terrane is close to Australia (Zhu et al., 2011, 2013). AM—Amuria; GI—Greater India; IC—Indochina terrane; LS—Lhasa terrane; NCB—North China block; NQ—North Qiangtang terrane; SCB—South China block; Si—Sibumasu terrane; SQ—South Qiangtang terrane; ZM—Zhongba microterrane.

tectonic terranes on the Tibetan Plateau and demonstrated that, before the Permian Sakmarian stage, the biological assemblages of these terranes belonged to the Gondwana bioregion. In different periods of the Permian, these terranes (such as the Lhasa terrane, South Qiangtang terrane, etc.) separated from the northern margin of East Gondwana (Zhang et al., 2013). Moreover, a mid- to high paleolatitude in the Southern Hemisphere can be inferred for the Tethyan Himalayas and Qiangtang terrane from the cold-water Gondwana-type biological assemblages, which are well documented by the glacial deposits on Gondwana (Caputo and Crowell, 1985). In addition, the tectonic model and paleogeographic reconstruction (Huang et al., 2018; Li et al., 2018; Zhao et al., 2018; Xian et al., 2019; Wang et al., 2021) also verified that these terranes were united as a whole, and they did not split from East Gondwana in the Early Devonian.

Due to the uncertainty of true polar wander (Goldreich and Toomre, 1969) and the longitudinal paleopositions of the microterranes, it is difficult to reconstruct Gondwana, especially before 320 Ma. A novel plate reconstruction method was provided by Torsvik et al. (2014). The surface-to-core-mantle boundary correlation was applied by these authors to constrain the longitude of the continent. Moreover, based on the true polar wander, they revised the paleomagnetic reconstruction frame. Finally, combining published detrital zircon U-Pb ages, paleomagnetic results, and paleontological data, we revised the Early–Middle Devonian paleogeographic position of the Zhongba microterrane and its surrounding terranes based on the late Paleozoic (408–388 Ma) results of the global plate reconstruction provided by Torsvik et al. (2014) and Matthews et al. (2016; Fig. 11). The reconstruction suggests that the Zhongba microterrane, the South Qiangtang terrane, the Tethyan Himalayas, and other terranes were integrated with, and located on the northern margin of, East Gondwana, at least in the Early–Middle Devonian.

CONCLUSIONS

We report for the first time paleomagnetic results from the Early–Middle Devonian Nadenger Formation of the Zhongba microterrane. The paleomagnetic results passed the fold and reversal tests, together with rock magnetism and petrographic observations, indicating that the results are reliable and therefore can be used for paleogeographic reconstructions. We constrained the paleogeographic location of the Zhongba microterrane in the Early and Middle Devonian to be 50.0°S ± 11.7°S. Based on detrital zircon U-Pb age spectra, published paleontological and paleomagnetic data, and plate-tectonic reconstruction results, we determined that the Zhongba microterrane likely belonged to the Tethyan Himalayas–Greater India–South Qiangtang tectonic system in the late Paleozoic, and we therefore present a revised paleogeographic pattern for the northern margin of Gondwana in the Devonian.

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et al., 2016). Cureval software (Hrouda, 1994) was used to present *k-T* curve features. Paleogeographic reconstructions were carried out using Gplate software (Williams et al., 2012), with data sourced from Matthews et al. (2016). Our paleomagnetic data are stored in Zenodo (https://doi.org/10.5281/zenodo.7854649), and the Supplemental Material[1] (Tables S1 and S4) can be accessed by everyone. Paleozoic detrital zircon data from the Zhongba microterrane were obtained from Zhang (2015) and Lin et al. (2020), and these data are listed in the Supplemental Material (Tables S2 and S3). Paleozoic detrital zircon data from the South Qiangtang terrane, Tethyan Himalayas, and Lhasa terrane were derived from Gao et al. (2022), and all the data can also be obtained from Zenodo (https://doi.org/10.5281/zenodo.7854649). References to the datasets and software are cited in the paper and listed in the References Cited.

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